The Interview-Compatible Recommender: A Measuring Instrument for Cold-Start Conversational Recommendation

Thesis Chapter A (v2)

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Abstract

Before one can study *how* to interview a cold-start user—which questions to ask, in what order—one needs a recommender that can *be* interviewed: a model that folds an arbitrary-length, mixed-channel set of answers into its belief and re-ranks a large catalogue, without per-user retraining and without trading away full-profile accuracy. We argue that this recommender is a *measuring instrument* for the entire elicitation-policy literature, and that its requirements are more demanding than any single published system meets. We state five requirements—**R1** state-of-the-art full-profile accuracy, **R2** arbitrary-length cold-start fold-in, **R3** channel-agnostic input (items, out-of-catalogue concepts, entities, and continuous directions through one interface), **R4** an uncertainty-native belief, and **R5** monotonicity (a truthful answer never lowers ranking quality)—and three acceptance gates **G0/G1/G2** that operationalise them. A survey of six model families shows the literature is *bifurcated*: the models that hit the full-profile bar (shallow linear/autoencoder recommenders such as EASE [20] and RecVAE [19]) are item-only point estimators, while the models with an uncertainty-native, multi-channel belief (soft-attribute Gaussian elicitation [3], unified item+attribute bandits [11], conjugate critique folds [23]) do not report full-profile accuracy at the R1 bar. We are aware of no published system that occupies the conjunction of all five. The instrument we propose to close this gap is a *frozen* certified shallow-SOTA tower with an analytic conjugate-Gaussian belief layer over its latent, into which every channel folds as a linear observation; three of its input legs—signed/graded values, out-of-catalogue concepts, and continuous direction tokens—are architecturally inexpressible in the item-indicator basis of the R1-bar models. This chapter contributes the requirement framework, the taxonomy and gap analysis, the instrument's design, and the experimental protocol (a metric-certification bridge to the published ML-20M numbers, then G0/G1/G2 on the ML-25M harness). Our two in-house anchors—a set-encoder tower at full/tail NDCG@10 0.4917/0.3071 and a RecVAE-class d=512 ruler at 0.4998/0.3443 (both ML-25M, not yet bridged to the published ML-20M scale)—are reported as such; all downstream instrument results are marked **[TODO:]** pending the certified run.

Contents

1 Introduction: the recommender as a measuring instrument

A *cold-start* conversational recommender must elicit taste from a short interview and then rank a large catalogue. Two questions are entangled: *what to ask* (the elicitation policy) and *how to turn answers into recommendations* (the recommender). The elicitation-policy literature is large and active, but almost every comparison in it is confounded: a policy's apparent value depends on the recommender it drives, and reported gains routinely conflate a better *question-selection strategy* with a better *answer-ingestion model.* Before we can measure elicitation strategies in isolation, we need to fix the recommender—and not just any recommender, but one strong and flexible enough that a good interview is not wasted on it. We call such a recommender an *interview-compatible instrument.*

Five requirements. An instrument fit for measuring conversational elicitation must simultaneously satisfy:

R1 — full-profile SOTA. On a full interaction history it must rank at the level of the strongest shallow collaborative-filtering baselines (EASE [20], RecVAE [19]), so that elicitation is measured against a recommender nobody can dismiss as weak [6].

R2 — any-length cold-start fold-in. It must ingest an evidence set of *any* cardinality, from 0 (fully cold) upward, without per-user retraining.

R3 — channel-agnostic input. Items, out-of-catalogue concepts, entities, and free text (and, for the continuous policy, direction tokens) must all fold through *one* interface as arbitrary embeddings, not a fixed categorical schema.

R4 — uncertainty-native belief. It must carry an explicit covariance over the user belief, not just a point estimate, so that a policy can act on *what it does not yet know.*

R5 — monotonicity. A truthful answer must never *lower* ranking quality; adding evidence should, in expectation, only help.

The bifurcation observation. The central empirical fact this chapter documents is that R1 and {R3, R4, R5} are *anti-correlated across the published literature.* The models that hit the full-profile bar—closed-form linear autoencoders (EASE [20], EDLAE [21]), amortised VAEs (Mult-VAE [12], RecVAE [19]), and training-free graph filters (GF-CF [18], BSPM [4], Turbo-CF [16])—are *item-only point estimators* on a one-hot indicator basis. The models with an uncertainty-native, multi-channel belief—soft-attribute Gaussian elicitation [3], item+attribute Thompson bandits [11], conjugate critique folds over a frozen factorisation [23], and per-item Bayesian NL elicitation [1]—do *not* report full-profile accuracy at the R1 bar. We call these the *accuracy corner* and the *belief corner*. No published system we are aware of occupies both (§4).

Contributions. This chapter is the instrument paper of the thesis. It does not claim a new elicitation policy (that is the companion method chapter); it claims the *recommender that makes policy measurement possible*. Concretely:

1. **The conjunction as the contribution.** We frame R1∧R2∧R3∧R4∧R5 as a single requirement no published system meets, and give a six-family taxonomy and per-near-miss gap analysis (§3, §4) that locates exactly which requirement each candidate fails. We say we are *aware of none*, never that there *is* none.
2. **Three architecturally-inexpressible legs.** We isolate the three input channels that are impossible (not merely untuned) in the item-indicator basis of every R1-bar model: (i) signed/graded per-entity values, (ii) out-of-catalogue concepts, and (iii) continuous direction tokens. These are the narrow, defensible novelty legs of the channel-agnostic interface (§4, §5).
3. **The instrument.** A design for a *frozen* certified shallow-SOTA tower with an analytic conjugate-Gaussian precision-accumulator belief layer over its latent, into which every channel folds as a linear observation (§5). The frozen-tower+exact-conjugacy shape is the one mechanism with a principled monotonicity story [23], and it is what separates us from the co-trained, approximate-posterior near-miss of Bıyık et al. [3].
4. **G0/G1/G2 as a reusable acceptance protocol.** We define three gates—full-strength preserved, posterior strictly shrinks per question, and cold-start per-question NDCG rises on *full and tail* versus random—and propose them as a portable certificate any interview-compatible recommender should have to pass (§2, §6).

We are careful throughout to distinguish what is *new* (the conjunction; the frozen-certified-tower wedge; the G0–G2 protocol on a SOTA tower) from what is *pre-empted* and must be cited (attributes-as-directions plus a Bayesian fold [3]; unified item+attribute belief [11]; conjugate folds over a frozen factorisation [23]; permutation-invariant set-encoders with information-gain acquisition [15]; value-carrying set-input cold-start recommenders [13]; the non-degradation-of-information principle, Good 1967 / Blackwell). "Uncertainty-native recommender" alone is a crowded 2024–25 area [2] and is *not* claimed as novel.

2 Requirements and gates

We now state the requirements and gates formally. Let the catalogue have n items and let the frozen tower expose a d-dimensional latent in which each item i has a factor row $\mathbf{Q}_i \in \mathbb{R}^d$. An *evidence set* after t questions is a set of observations $\mathcal{E}_t = \{(\boldsymbol{\phi}_j, y_j)\}_{j=1}^{t}$, where $\boldsymbol{\phi}_j \in \mathbb{R}^d$ is the latent direction of the queried entity (an item factor, a concept centroid, an entity embedding, or a continuous direction token) and $y_j \in \mathbb{R}$ is the (possibly graded, possibly signed) answer. The instrument maintains a belief $b(\mathcal{E}_t) = (\boldsymbol{\mu}_t, \boldsymbol{\Sigma}_t)$ over the user latent $\mathbf{u} \in \mathbb{R}^d$ and scores item i by a fixed read-out $s_i(\boldsymbol{\mu}_t)$ of the base tower.

Requirements (formal).

R1. With $\mathcal{E}$ equal to a user's full history, NDCG@10$\big(s(\boldsymbol{\mu}(\mathcal{E}))\big)$ ties the strongest shallow baseline on a shared harness (Def. 1).

R2. $b(\mathcal{E}_t)$ is defined and computable for every $t \geq 0$ with no gradient step at inference; t=0 yields the prior belief (a non-personalised ranker).

R3. The update from $\mathcal{E}_t$ to $\mathcal{E}_{t+1}$ depends on the new observation only through the pair $(\boldsymbol{\phi}_{t+1}, y_{t+1}) \in \mathbb{R}^d \times \mathbb{R}$; the operator is identical for every channel, so any entity with an embedding is a legal observation—including entities absent from the catalogue.

R4. $\boldsymbol{\Sigma}_t$ is an explicit, queryable covariance; a policy may read $\boldsymbol{\Sigma}_t$ to choose the next question.

R5. For a truthful answer y_{t+1}, $\mathbb{E}\big[\mathrm{NDCG@10}(s(\boldsymbol{\mu}_{t+1}))\big] \geq \mathrm{NDCG@10}(s(\boldsymbol{\mu}_t))$. We treat R5 as an *empirical* property to be demonstrated per step, not a theorem for a learned ranker (§7).

Definition 1 (Ties SOTA)**.** *A recommender* ties *a baseline on a metric if their scores are within the paired per-user bootstrap confidence interval on one shared harness (same users, items, split, metric depth). A number computed on a different metric/split (e.g. NDCG@10 on ML-25M vs. NDCG@100 on the ML-20M strong-generalisation split) is* not *a tie until bridged (§6.2).*

Gates (acceptance protocol). The instrument is accepted only if it passes all three:

G0 — full strength preserved. Evaluating on the belief mean with the full profile reproduces the base tower's full-profile NDCG@10 (full *and* tail), i.e. the belief layer adds no full-profile regression. This certifies R1 survives the addition of the belief machinery.

G1 — usable posterior. The posterior strictly shrinks per truthful question: $\det \boldsymbol{\Sigma}_{t+1} < \det \boldsymbol{\Sigma}_t$ (and the shrinkage is directed along the queried $\boldsymbol{\phi}_{t+1}$). This certifies R4.

G2 — cold-start curve rises on full and tail. Starting cold (t=0), per-question NDCG@10 rises with t on *both* the full catalogue and the long tail, and beats a random-question control at every budget with a paired per-user bootstrap CI excluding zero. This is the operational form of R2$\wedge$R5 and the headline instrument claim.

We report *both* full and tail NDCG@10 for every gate (a project reporting rule: full-only or tail-only hides the effect, which historically has been much larger on the tail than on the full catalogue). All gate results in this chapter are **[TODO:]** pending the certified run of §6.

3 Related work: a six-family taxonomy

We organise interview-relevant recommenders into six families and score each against R1–R5 (Table 1). The reading is consistent across families: no family is strong on both the accuracy axis (R1) and the belief-channel axis (R3–R5). Figure 1 plots the same fact as a qualitative gap map. Table 1 is the compact conceptual overview; the chapter's anchor is the per-system *master table* (Table 2), which classifies every reasonable candidate against R1–R5 *and*, where one exists, prints its full/tail NDCG@10 on our shared ML-25M ruler—so that the bifurcation is not merely asserted from published claims but is visible in the one column of measured numbers.

The one column that is blank for an entire block. Read Table 2 down its measured-accuracy column. Every system that carries a number—a full/tail NDCG@10 on our shared ML-25M ruler—sits in block (a) or (b): the item-only linear and amortised recommenders. The entire belief corner, block (c)—EDDI, Golbandi, RBMF/Functional-MF, Bıyık, ConTS, BCIE, PEBOL, the LLM-zero-shot CRS—has a *blank* accuracy column: not one of these systems has a full-catalogue top-N number on any shared ruler, and its R1 cell is ✗ in every row. This is not an accident of our harness. Most CRS and elicitation papers never report full-catalogue top-N accuracy at all (they report elicitation RMSE, cold-start MRR/MAP on a shortlist, or critique metrics), and several *cannot produce it by construction*: a per-item Beta posterior (PEBOL) or a decision tree over rating queries (Golbandi) does not induce a ranking over the whole catalogue, and an LLM CRS needs an API call per candidate to score one. Conversely, every system with a competitive number is item-only and point-estimate—✗ on R3 and R4. The two blocks are disjoint: no row combines a ✓ (or even a filled-in number) in the accuracy column with ✓s across R3–R5. The top-right cell of the gap map—all five requirements met *and* a competitive full-profile number—is occupied only by the target row, which is a design, not a result. That empty conjunction is the chapter's thesis, and the master table is where it is impossible to miss.

The accuracy corner (families ii–iii). On the canonical ML-20M strong-generalisation split (10k validation / 10k test held-out users, NDCG@100), the full-profile frontier has been *flat since 2019–2020* and is held by shallow models, not deep nets: EASE [20] at NDCG@100 0.420, Mult-VAE [12] at 0.426, RecVAE [19] at 0.442 (top of this split), with EDLAE [21] in the same class; SLIM (0.401) and WMF/iALS (0.386) are the classic floors. Dacrema et al. [6] is the binding warning: most deeper "SOTA" gains vanish on this exact split. The graph-filter leaders (GF-CF [18], BSPM [4], Turbo-CF [16]) are the 2023–24 speed+accuracy front on Gowalla/Yelp/Amazon but do not report this ML-20M split, so they are admissible to our accuracy tier only if re-run on our harness. All of family (ii)/(iii) are item-only: their input is a one-hot (or set-indicator) interaction vector, which *cannot encode* a signed value, an out-of-catalogue concept, or a continuous direction—R3 is an architecture property here, not a tuning gap.

The belief corner (family iv, and vi). Bıyık et al. [3] maintain a multivariate-Gaussian belief over a user embedding and treat both item comparisons and "soft attributes" (e.g. "scarier") as directions (concept-activation vectors) in the item space, selecting queries by expected value of information—the nearest thing to R2+R3+R4 together. ConTS [11] unifies items and attributes as Thompson-sampling arms with a posterior over a cold-start user. BCIE [23] folds critiques into a *frozen* factorisation by exact conjugate-Gaussian updates—the closest published analogue of our belief mechanism. PEBOL [1] keeps a per-item Beta posterior updated by LLM-extracted aspects and NLI scoring, reporting cold-start MRR@10 0.27 versus a 0.17 baseline. None of these reports full-profile accuracy at the EASE/RecVAE bar; each is a belief layer, not a certified tower.

The set-encoder and cold-start-NP precedents (families i, iii). EDDI/Partial-VAE [15] is the permutation-invariant set-encoder over an arbitrary observed subset with information-gain acquisition—exactly R2's architecture, but demonstrated outside recommendation and without a full-profile CF result or an attribute/concept channel. TaNP [13] gives a neural-process, stochastic-process uncertainty over a value-carrying set input for cold-start—the R2+R4 shape, but not at full-profile SOTA nor channel-agnostic. These are cited as the mechanism precedents our design instantiates, not as rivals for the conjunction.

The gap map. Figure 1 places the families on two qualitative axes: *interview-compatibility* (channels + uncertainty + monotonicity, i.e. R3–R5) on the x-axis, and *full-profile accuracy* (R1)

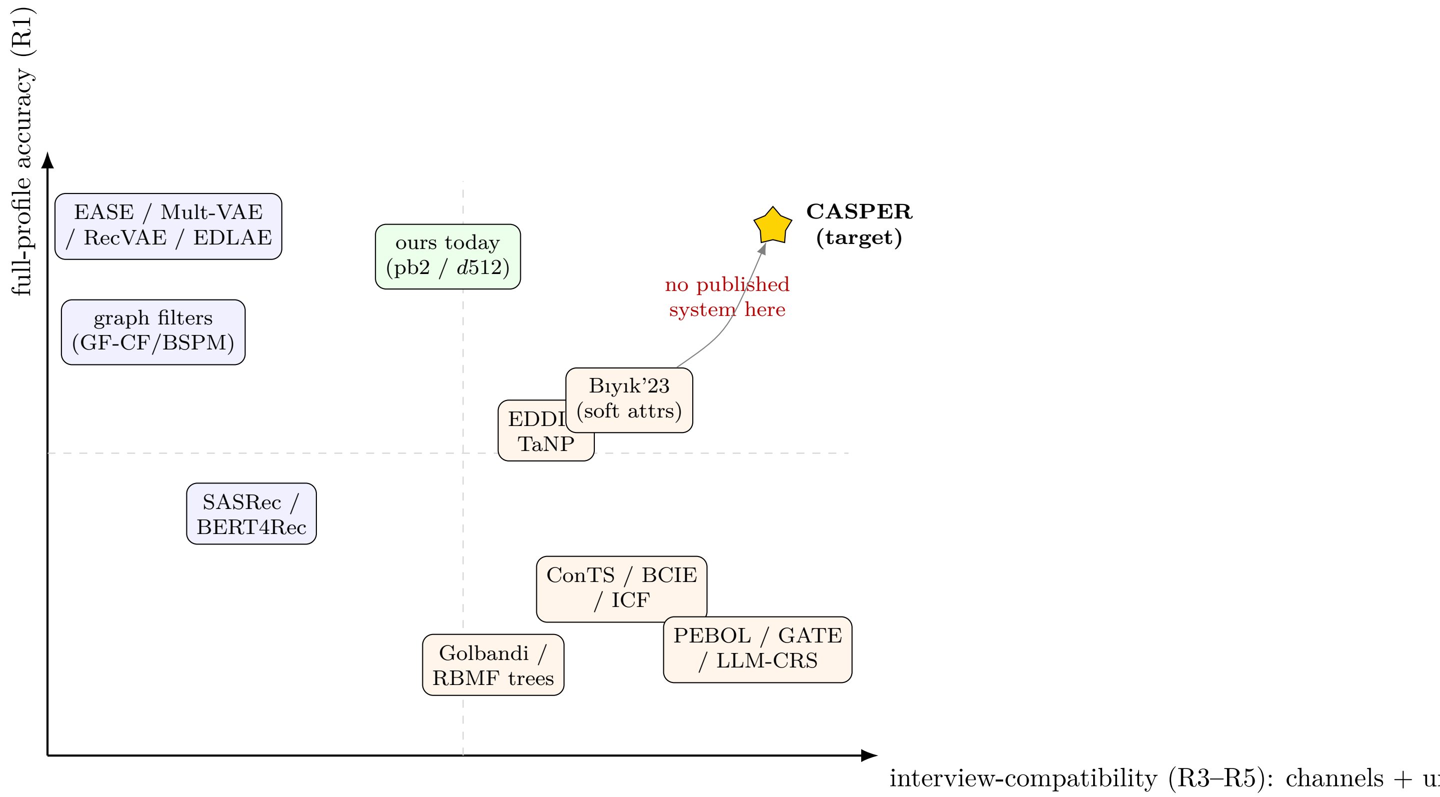

Figure 1: **The gap map (qualitative).** x-axis: interview-compatibility (channel-agnostic input, an uncertainty-native belief, and monotonicity—R3–R5). y-axis: full-profile accuracy (R1). Positions are *qualitative* and reflect our reading of each system's published claims, not measured coordinates; they are meant to show the *shape* of the bifurcation, not exact rankings. The accuracy corner (EASE/RecVAE/EDLAE [20, 19, 21], graph filters [18, 4], sequence models [9, 22]) is top-left; the belief corner (EDDI [15]/TaNP [13], Bıyık [3], ConTS [11]/BCIE [23]/PEBOL [1]) is lower-right. The top-right target is empty: we are aware of no published system there.

on the y-axis. The accuracy-corner systems sit top-left; the belief-corner systems sit lower-right; the target sits top-right, empty.

4 Gap analysis

The verdict. No single published system satisfies all of {R1 SOTA-full-profile ∧ R2 any-length set ∧ R3 channel-agnostic via arbitrary embeddings ∧ R4 uncertainty-native ∧ R5 monotone}. The binding near-miss is Bıyık et al. [3]: it fails R1 (a co-trained item space, not a frozen certified RecVAE-class tower, and no full-profile benchmark) and gives only an approximate, "generally not Gaussian" posterior; every other candidate fails R3 or R4 outright. **We are aware of none** that occupies the conjunction—and we state it exactly that way, never as "there is none."

Per-near-miss failure analysis.

- **Bıyık et al. 2023** [3] — has R2 + R3 (items and soft attributes as directions) + R4 (Gaussian belief). *Fails R1*: the item space is co-trained with the belief model, not a certified full-profile

CF tower, and accuracy is reported on their elicitation task, not the EASE/RecVAE bar. R3 is *attribute directions*, not arbitrary embeddings / open concepts / continuous query tokens. R5 is approximate (sampled EVOI over a non-Gaussian posterior). This is the central threat, and our wedge is exactly (a) frozen certified tower vs. co-trained, (b) exact conjugacy vs. approximate posterior, (c) arbitrary / open / continuous tokens vs. a fixed attribute set.

- **ConTS** [11] — R2 + R4 + a unified item/attribute action space. *Fails R3* (a fixed categorical attribute schema; no arbitrary embedding, no continuous or open token) and *R1* (a bandit over arms, not a full-profile encoder).
- **EDDI / Partial-VAE** [15] — R2 + R4 (information-gain) + set input. *Fails R1* (not a recommendation model / no CF SOTA) and *R3* (no attribute or concept channel as demonstrated).
- **RecVAE / EASE** [19, 20] — *R1 yes. Fail R3* (an item-only one-hot / indicator basis—cannot express a concept, a signed value, or a continuous direction) and *R4* (point estimate; RecVAE's latent posterior is item-only and is not used as an actionable belief).
- **PEBOL** [1] — R4 (per-item Beta) + R3 (LLM aspects) + strong cold-start. *Fails R1* (no full-profile CF ranking) and its R4 is *per item*, not a shared user covariance; no monotonicity guarantee.
- **SASRec / BERT4Rec** [9, 22] — R2 (re-encode the sequence). *Fail R3, R4, R5* (item-only, order-sensitive, point estimate, non-monotone—a later token can reorder everything). Token input is trivially satisfied but is *not* value-carrying or mixed-type, so "interview = tokens" does not hold; and BERT4Rec's fold-in advantage carries a replicability caveat [17].

The three inexpressible legs. Three input channels are *architecturally impossible* (not merely untuned) in the item-indicator basis of every R1-bar model: (i) graded/signed per-entity values (the basis has no sign and no magnitude—a "dislike" is not representable as a negative one-hot without changing the likelihood), (ii) out-of-catalogue concepts (there is no column for an entity absent from the catalogue), and (iii) arbitrary continuous direction tokens. Folding these as *linear observations in a frozen latent* is the one leg of the conjunction we find nowhere in the literature.

Pre-empted—cited, never claimed as first. We must credit, and do not claim priority over: attribute-answers-as-directions plus a Bayesian fold [3]; unified item+attribute belief and value-of-information [3, 11]; conjugate-Gaussian folds over a frozen factorisation [23]; permutation-invariant set-encoders with information-gain acquisition [15]; value-carrying set-input cold-start recommenders [13]; the non-degradation-of-information principle (Good 1967 / Blackwell); and the now-crowded "uncertainty-native recommender" area [2]. Our defensible novelty is only the *conjunction* of all five requirements in one frozen certified tower, plus the three inexpressible legs and the G0–G2 protocol.

5 The instrument

The design follows directly from the gap analysis: use the shallow frozen tower that already holds the accuracy corner (R1, and the flat-since-2019 frontier means no deep-net detour is warranted), and add the one belief mechanism with a principled monotonicity story—exact linear-Gaussian conjugacy—over its frozen latent (R4, R5), with a channel-agnostic linear-observation interface (R3). This section is written at design level; implementation details and all numbers are **[TODO:]**.

5.1 Frozen shallow-SOTA tower

The substrate is a certified shallow recommender—a RecVAE-class amortised encoder or a closed-form autoencoder [19, 20]—trained once on the full catalogue and then *frozen*. It exposes (a) item factor rows $\mathbf{Q}_i \in \mathbb{R}^d$, (b) a decoder read-out $s_i(\mathbf{u}) = \mathbf{Q}_i^\top \mathbf{u} + \beta_i$ with a learned bias β_i, and (c) the encoder that maps a full interaction set to a latent. Freezing is the wedge against Bıyık [3] (co-trained) and the guarantee that G0 (full strength preserved) is a property of the substrate, not of the belief layer. We deliberately do *not* extend the linear basis with concept columns (that stays in the item-indicator world and fails R3), nor learn the uncertainty head (no monotonicity story, and retraining risks G0). [**TODO:** Select and certify the frozen tower (RecVAE-$d512$ ruler candidate; report the certified full/tail NDCG@10 and the bridge number of §6.2).]

5.2 Analytic conjugate-Gaussian belief layer

Over the frozen latent we place a Gaussian belief on the user vector, $\mathbf{u} \sim \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$, and model each answer as a noisy linear read-out along the queried direction:

$$y_j \; = \; \boldsymbol{\phi}_j^\top \mathbf{u} + \varepsilon_j, \qquad \varepsilon_j \sim \mathcal{N}(0, \sigma_j^2). \tag{1}$$

Because the observation is linear-Gaussian, the posterior after $\mathcal{E}_t$ is Gaussian in closed form; maintaining it in *precision* form (a precision accumulator) makes the any-length fold-in a single running sum:

$$\boldsymbol{\Lambda}_t \; = \; \boldsymbol{\Lambda}_0 + \sum_{j=1}^{t} \sigma_j^{-2}\, \boldsymbol{\phi}_j \boldsymbol{\phi}_j^\top, \qquad \boldsymbol{\Lambda}_t\, \boldsymbol{\mu}_t \; = \; \boldsymbol{\Lambda}_0 \boldsymbol{\mu}_0 + \sum_{j=1}^{t} \sigma_j^{-2}\, y_j\, \boldsymbol{\phi}_j, \qquad \boldsymbol{\Sigma}_t = \boldsymbol{\Lambda}_t^{-1}. \tag{2}$$

Equation (2) is exactly the mechanism BCIE [23] uses for critiques over a frozen factorisation; our contribution is doing it over a *certified full-profile SOTA* tower with channel-agnostic tokens, and certifying the G0–G2 protocol on it. Three properties fall out by construction: it is a permutation-invariant *set* operation defined for every $t \geq 0$ (R2); each update adds a rank-1 term along $\boldsymbol{\phi}_{t+1}$, so $\boldsymbol{\Lambda}_t$ only grows and $\det \boldsymbol{\Sigma}_t$ only shrinks (G1); and conjugate updates are Blackwell/Good-1967 monotone in *expected utility*. The last is why R5 is plausible, but—because the tower's dot-product ranker does not act optimally under the belief—per-question NDCG monotonicity is *not* automatic and must be shown empirically (§7). [**TODO:** Fix $\boldsymbol{\Lambda}_0, \boldsymbol{\mu}_0$ (prior from the tower), the per-channel noise σ_j, and the confidence/temperature schedule; verify G0 holds at the belief mean on the full profile.]

5.3 Channel-agnostic tokens as linear observations

Every channel supplies a direction $\boldsymbol{\phi}_j \in \mathbb{R}^d$ and a value $y_j \in \mathbb{R}$ to Eq. (2); the update is identical across channels (R3):

- **Items.** $\boldsymbol{\phi}_j = \mathbf{Q}_i$ (the frozen item factor). A graded or signed answer sets y_j; a dislike is a negative y_j along the same direction—representable here precisely because the observation model (1) has a sign and a magnitude, unlike a one-hot indicator.
- **Out-of-catalogue concepts.** $\boldsymbol{\phi}_j$ is a concept direction in the latent (e.g. a centroid of member factors, or a text-adapter image of the concept), folded exactly like an item—no catalogue column required.
- **Entities / free text.** An embedding is projected into the latent by a fixed adapter and folded identically.

- **Continuous direction tokens.** A policy may emit any $\boldsymbol{\phi}_j \in \mathbb{R}^d$ directly (the continuous-action interface the companion chapter uses); the belief update does not distinguish it from a catalogue entity.

The signed/graded, out-of-catalogue, and continuous legs are the three channels inexpressible in the R1-bar basis (§4). [**TODO:** Specify the concept-direction and text-adapter constructions, the polarity/gradation encoding, and the acceptance tests (a like/dislike flip reverses the queried neighbourhood; a concept fold lifts its member items; monotone in t).]

6 Experimental design

6.1 Baseline bank

The baseline bank (Table 3) is three tiers: classic floors that every curve must clear (Dacrema et al. [6] demands them), the full-profile SOTA bar that decides G0, and the elicitation-compatible rivals that decide G1/G2. Reported anchors are on the systems' *own* splits and are not comparable to ours until bridged (§6.2); we print them only to show what "ties SOTA" will mean.

6.2 Evaluation protocol

(i) Certification bridge. Our in-house anchors are on a *different* ruler than the published bar, and we do not conflate them. Concretely, our set-encoder tower reaches full/tail NDCG@10 0.4917/0.3071 and our RecVAE-class d=512 ruler reaches 0.4998/0.3443, *both on ML-25M at NDCG@10*—whereas the published EASE/RecVAE numbers are NDCG@100 on the ML-20M strong-generalisation split. These are not comparable, and no "ties SOTA" claim is made until the bridge is run. The bridge has two steps: (1) *reproduce* the published EASE and Mult-VAE/RecVAE numbers on the ML-20M Liang split (NDCG@100, Recall@20/50) in our own code, snapping to the reported values exactly [20, 12, 19]; (2) *port* the certified tower to the ML-25M harness and report full/tail NDCG@10 there. We keep ML-20M in the loop despite being older because the Liang 2018 split is a protocol lock-in the whole full-profile literature reports on; ML-25M is the newer corpus and, crucially, carries the genome tags the concept channel needs, so both datasets are required—one for comparability, one for the channels.

The first step is complete. On the ML-20M Liang strong-generalisation split our split statistics reproduce the published counts exactly (9,990,682 events / 116,677 train users / 20,108 items), and both anchor implementations **snap to their published values**: EASE at NDCG@100 0.4203 vs. the reported 0.420 [20], and RecVAE at 0.4425 vs. the reported 0.442, with Recall@20 (0.4144 vs. 0.414) and Recall@50 (0.5525 vs. 0.553) matching as well [19] (Mult-VAE/DAE certify the same way; [**TODO:** fill Mult-VAE/DAE snap deltas]). One closed-form and one neural reproduction to the third decimal certify that our split construction, metric implementation, and both implementation families are faithful to the published protocol—every ML-25M baseline number below therefore comes from certified code, and we spend no further space on it.

(ii) G0 — full-profile on ML-25M, items-only. On the baselines' native input (items only), evaluate full and tail NDCG@10 on the full ML-25M catalogue with a paired per-user bootstrap CI, confirming the instrument's belief mean reproduces the frozen tower and ties EASE/RecVAE on the shared harness. The item-only baselines are measured (Table 4).

The closed-form linear models set the G0 bar, and we report it as such. On our ML-25M harness EDLAE attains 0.5230 full / 0.3464 tail and EASE 0.5078/0.3394—both *above* our interview-native encoder (0.4917/0.3071) and our RecVAE-class ruler (0.4998/0.3443). All rows share the same split, cohort, tail definition, seed set, and NDCG@10 routine (verified by code inspection). The instrument's R1 obligation is therefore to close a $\sim$ 3-point full-profile gap to EDLAE (0.5230; EASE 0.5078), not merely to tie RecVAE. We report this rather than excluding the stronger baselines.

(iii) G2 — per-question curves. Cold-start per-question NDCG@10 curves (full and tail) versus a random-question control. Items are foldable by all baselines; mixed channels (concepts, signed values, continuous tokens) are foldable only by our instrument, which is the point of the comparison—the rivals cap out at what an item-only interview can carry. Symmetric access is enforced where a baseline *can* fold a channel; where it structurally cannot, that is reported as an architecture limit, not hidden. [**TODO:** G2 curves for items-only (all methods) and mixed-channel (ours), full & tail, with CIs and the random control; G1 posterior-shrinkage diagnostic.]

(iv) Concept-as-pseudo-item ablation. To pre-empt "just add genre columns," we fold a concept into EASE/RecVAE as the centroid-of-members pseudo-item and compare against our concept channel. The expected finding is that a pseudo-item column is a lossy special case (it cannot carry sign, magnitude, or an out-of-catalogue direction), but we run it rather than assert it. [**TODO:** Ablation table: concept-as-pseudo-item fold into EASE/RecVAE vs. our concept channel, full & tail.]

Metrics. Full *and* tail NDCG@10 on ML-25M (the project reporting rule, scored with the decoder's learned bias, not a log-count popularity floor), plus NDCG@100 (and Recall@20/50) on the ML-20M bridge split for comparability with the published anchors. Every headline delta carries a paired per-user bootstrap CI on one shared harness. All result cells are [**TODO:**].

7 Limitations and threats to validity

- **Metric non-comparability until the bridge is run.** Our 0.4917/0.3071 and 0.4998/0.3443 are NDCG@10 on ML-25M; the published bar is NDCG@100 on ML-20M. Until §6.2 snaps our code to the published numbers, *no* "ties SOTA" claim is licensed—the numbers live on different rulers.
- **Monotonicity is empirical, not a theorem.** Conjugate-Gaussian updates are monotone in expected utility (Good 1967 / Blackwell) *only in expectation*, and only if the ranker acts optimally under the belief—which a fixed dot-product / NDCG ranker does not. R5 (and G2's rising curve) is therefore a *measured* property with a per-question CI, not a guarantee; a truthful answer can in principle still lower NDCG at a given step, and we report any such step rather than suppress it.
- **Simulator-based evaluation pending a human study.** G1/G2 are measured under a stated answer model (a simulator), which shares structure with the instrument's own latent; this risks circularity (the answer, the belief, and the metric are all functions of the same space). We add answer-noise ablations and a simulator-alignment limitation to every gate, and treat a $\geq$1 human round-trip study as the load-bearing external validation—not yet run.

- **BERT4Rec replicability.** The sequence baseline's fold-in advantage is subject to a documented replicability caveat [17]; we tune it to the published protocol and flag any residual gap rather than treat a single run as authoritative.
- **Graph-filter admissibility.** GF-CF/BSPM/Turbo-CF [18, 4, 16] do not publish the ML-20M strong-generalisation NDCG; they enter Tier 2 only if re-run on our split, and are otherwise reported as an accuracy ceiling on other datasets, not as a bridged tie.
- **Absence of evidence.** The conjunction claim is "we are aware of none," bounded by our survey; a set-encoder / deep-set recommender that hits RecVAE-class full-profile accuracy *and* takes value-carrying mixed-type tokens would partly pre-empt R1∧R2∧R3. Our searches found no such reported system, which is absence of evidence, not proof of absence.

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Family	Representatives	R1	R2	R3	R4	R5	One-line verdict
(i) retrain / meta-learn per user	MeLU, TaNP [13], CDRNP	$\sim$	✓	✗	$\sim$	✗	R2/R4 precedent; not full-profile SOTA, not channel-open.
(ii) fold-in linear / closed-form	EASE [20], SLIM, GF-CF [18], BSPM [4], Turbo-CF [16]	✓	✓	✗	✗	$\sim$	The full-profile bar (G0); item-indicator basis $\Rightarrow$ R3 impossible.
(iii) amortised inference encoders	Mult-VAE [12], RecVAE [19], EDLAE [21], EDDI [15]	✓	✓	$\sim$	$\sim$	✗	The architecture we live in; VAEs item-only, EDDI right shape/wrong domain.
(iv) Bayesian belief over item space	Bıyık [3], ConTS [11], BCIE [23], ICF [24]	✗	✓	$\sim$	✓	$\sim$	The R4/R5 heart; belief layers, not full-profile towers.
(v) sequence models	SASRec [9], BERT4Rec [22]	$\sim$	✓	✗	✗	✗	Interview $\neq$ ordered session; tokens item-only, order-sensitive.
(vi) LLM-as-recommender / CRS	PEBOL [1], GATE [10], LLM-CRS [8]	✗	✓	✓	$\sim$	✗	Best R3 (open text); weakest R1 (weak collaborative signal).
CASPER (target)	**this chapter**	✓	✓	✓	✓	✓	**The conjunction; no published system occupies it.**

Table 1: Six-family taxonomy scored against R1–R5. ✓ met, $\sim$ partial, ✗ not met. Scores are our reading of each family's *published* claims and are qualitative; the last row is a target, not a result. R1 and {R3,R4,R5} are anti-correlated: families (ii)/(iii) hold the accuracy corner, family (iv) the belief corner, and none spans both. Adapted from the landscape analysis (c2).

						(our ML-25M ruler)		(own benchmark, *NOT* comparable)
(a) Classic / linear full-profile								
Most-Popular	✗	✓	✗	✗	~	.2851 / .0589	measured	n/r (non-personalised floor)
Item-kNN (cosine)	✗	✓	✗	✗	~	.4272 / .2208	measured	@100 ~ 0.36 class [5]
PureSVD / iALS-WMF	✗	✓	✗	✗	~	.4272 / .3172	measured (iALS)	@100 0.386 (WMF, ML-20M)
SLIM	~	✓	✗	✗	~	—	not-run	@100 0.401 (ML-20M)
EASE [20]	✓	✓	✗	✗	~	**.5078** / **.3394**	measured	@100 0.420 (ours **PASS** +.0003)
EDLAE [21]	✓	✓	✗	✗	~	**.5230** / **.3464**	measured	@100 ≈ 0.42–0.44 class
GF-CF / BSPM / Turbo-CF [18, 4, 16]	~	✓	✗	✗	~	—	not-run	n/r (ML-20M not reported)
(b) Amortized / neural full-profile								
Mult-DAE [12]	✓	✓	✗	✗	✗	[TODO:]	queued	@100 0.419
Mult-VAE [12]	✓	✓	✗	~	✗	[TODO:]	queued	@100 0.426
RecVAE [19] (d=512)	✓	✓	✗	~	✗	0.4998 / 0.3443	measured	@100 0.442 (ours **PASS** +.0005)
SASRec / BERT4Rec [9, 22]	~	✓	✗	✗	✗	—	not-run	n/r (next-item; repl. [17])
TaNP [13] (MeLU / CDRNP line)	~	✓	✗	✓	✗	—	not-run	n/r (own cold-start benchmark)
(c) Elicitation / CRS systems — the belief corner								
EDDI / Partial-VAE [15]	✗	✓	~	✓	✗	—	cannot-rank-full	n/r (non-recsys; UCI/MNIST)
Golbandi tree [7]	✗	✓	✗	✗	✗	—	not-run	n/r (Netflix RMSE)
RBMF / Functional-MF [14, 25]	✗	✓	✗	✗	✗	—	not-run	n/r (elicitation RMSE)
Bıyık soft-attributes [3]	✗	✓	~	✓	~	—	not-run	n/r (own elicitation task)
ICF [24]	✗	✓	✗	✓	~	—	not-run	n/r (online CF; own protocol)
ConTS [11]	✗	✓	~	✓	~	—	not-run	n/r (cold-start CRS metrics)
BCIE [23]	✗	✓	~	✓	~	—	not-run	n/r (critique metrics)
PEBOL [1]	✗	✓	✓	~	✗	—	needs-LLM	MRR@10 0.27 vs 0.17 (cold-start only)
GATE [10]	✗	✓	✓	✗	✗	—	needs-LLM	n/r (elicitation-only; no full ranking)
LLM zero-shot CRS [8]	✗	✓	✓	✗	✗	—	needs-LLM	n/r (weak collaborative signal)
(d) Ours								
Set-encoder tower (pb2)	~	✓	✗	✗	~	.4917 / .3071	measured	n/a (in-house)
RecVAE-class d=512 ruler	~	✓	✗	~	✗	.4998 / .3443	measured	n/a (in-house; ML-20M snap **PASS** 0.442)
Instrument (target)	✓	✓	✓	✓	✓	[TODO:]	pending	n/a (this chapter)

Table 2: **Master table: every reasonable system by R1–R5 and by measured full-profile accuracy on one shared ruler.** ✓ met, ~ partial, ✗ not met (our reading of each system's *published* claims, consistent with Table 1). *full/tail NDCG@10* is on *our* ML-25M half-split harness (seed-averaged $\{1, 2, 3, 7, 11\}$; test cohort = held-out users minus the 300 quarantined study users), scored with the decoder's learned bias. Status: *measured* (run on this ruler); *queued* ([TODO:], run pending); or "—" with a one-word reason (*not-run*, *cannot-rank-full*, *needs-LLM*). The final column is each system's own published full-profile anchor on its own split/metric—*not* comparable to our ruler until bridged (§6.2); "n/r" where the literature reports none. The RecVAE row is our own d=512 training run on this ruler; the "ours" block repeats it as the ruler row. Omitted systems: UNICORN-line graph-RL CRS (no full-profile claim; covered by the taxonomy discussion, §4); MeLU/CDRNP (folded into the TaNP row as one meta-learning line). No number is invented: a value

Baseline	Type	Reported anchor	Why the slot
Tier 1 — classic floors (must clear)			
Most-Popular	non-personalised	popb floor (our harness)	the "did we learn anything" line; G2 must beat it cold.
Item-kNN (cosine)	neighbourhood	ML-20M NDCG@100 ~ 0.36 class	instant fold-in; Cremonesi-era strong-simple [5].
PureSVD / iALS-WMF	linear MF fold-in	WMF ML-20M NDCG@100 0.386	cheap fold-in reference.
SLIM	sparse linear item-item	ML-20M NDCG@100 0.401	direct EASE ancestor.
Tier 2 — full-profile SOTA (the R1 / G0 bar)			
EASE [20]	closed-form item-item	NDCG@100 0.420	the bar to tie; item-only shows the R3 gap.
RecVAE [19]	amortised VAE	NDCG@100 0.442	top of the ML-20M split; our $d512$ ruler class.
Mult-VAE / DAE [12]	VAE / DAE	NDCG@100 0.426 / 0.419	canonical amortised anchor.
EDLAE [21]	denoising higher-order EASE	$\approx$ RecVAE class	confirms the flat frontier.
GF-CF / BSPM / Turbo-CF [18, 4, 16]	closed-form graph filter	SOTA on Gowalla/Yelp (ML-20M not reported)	admit only if re-run on our split.
SASRec / BERT4Rec [9, 22]	self-attentive sequence	strong next-item (not this split)	sequence fold-in; watch replicability [17].
Tier 3 — elicitation-compatible rivals (decide G1/G2)			
EDDI / Partial-VAE [15]	set-encoder + info-gain	non-recsys (UCI/MNIST)	the R2 architecture; myopic info-gain baseline.
Bıyık soft-attributes [3]	Gaussian belief + EVOI	own elicitation task	the central threat; frozen-vs-co-trained wedge.
ConTS [11]	Thompson bandit, item+attr arms	cold-start CRS metrics	unified-space + uncertainty; fixed schema.
BCIE [23]	conjugate-Gaussian critique fold	critique metrics	exact-conjugacy-over-frozen precedent.
PEBOL [1]	Bayesian-opt NL-PE (LLM)	MRR@10 0.27 vs 0.17	load-bearing PE baseline; per-item Beta.
Golbandi tree [7]	RMSE decision-tree interview	Netflix RMSE	the static-tree interview comparator.
RBMF / Functional-MF [14, 25]	MF-embedding seeds	elicitation RMSE	interview-seed line; cite-and-differentiate.

Table 3: Baseline bank. Anchors are on each system's own split/metric and are *not* comparable to ours until the bridge (§6.2) is run; G0 is decided against EASE+RecVAE, G1/G2 against the Tier-3 rivals plus a random-question control. Any deep-net "SOTA" not snapping to Tier-2 numbers on our split is inadmissible [6].

Method	full NDCG@10	tail NDCG@10	note
Most-Popular	0.2851	0.0589	train-likes popularity
Item-kNN (cosine)	0.4272	0.2208	neighbourhood fold-in
iALS-WMF	0.4272	0.3172	MF fold-in
EASE [20]	**0.5078**	**0.3394**	full@100 0.5136
EDLAE [21]	**0.5230**	**0.3464**	full@100 0.5297; fixed p=0.5
Ours			
Set-encoder tower (pb2)	0.4917	0.3071	interview-native encoder
RecVAE-class d=512 ruler	0.4998	0.3443	our in-house ruler

Table 4: G0 — full-profile, items-only, ML-25M half-split, NDCG@10 (full and tail), seed-averaged over $\{1, 2, 3, 7, 11\}$; test cohort = held-out users minus the 300 quarantined study users. The baseline evaluate path reuses the canonical harness (split/cohort/tail/seed protocol) verbatim, so the baselines are on the same ruler as our recorded anchors.